1. Probability Distribution and Hypothesis Testing

1.1 Probability Distribution

From the given stats, we want to find the probability of randomly selected fruits that will have blackspot at most 10 out of 100 randomly. Here we are going to use binomial distribution to calculate the probability because for each trial the probability of blackspots (Success) stays the same and the trials are independent, so the outcome of one trail does not affect the outcome of any other trial.

To find the probability of blackspots,

$$\textbf{Probability} = \frac{NH4\ with\ Blackspots}{Sample\ Size}$$

$$= \frac{14}{81} = 0.17$$

For calculating Binominal distribution, we use

BINOM.DIST(number_s, trials, probability_s, cumulative)

Where S signifies the Success. In this case success is blackspots.

BINOM.DIST (10, 100, 0.17, 1)

= 0.035

Therefore, probability of success for at most 10 out of 100 randomly selected fruits that have blackspots is found to be 0.035.

Fig 1. shows the Binominal Distribution curve for the above-mentioned problem.

Number of Trials	Binomial Distribution
0	0.00000001
1	0.00000017
2	0.00000185
3	0.00001309
4	0.00006890
5	0.00028838
6	0.00100013
7	0.00295776
8	0.00761892
9	0.01737803
10	0.03556759

Table 1

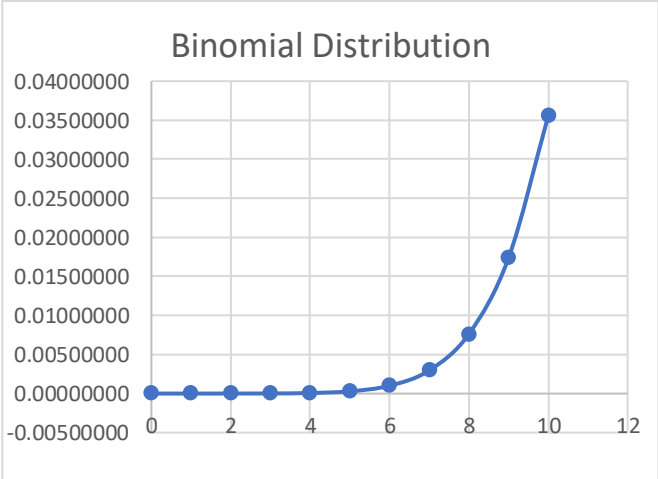

Fig 1. Binominal Distribution Curve

Judgment: The probability of Blackspots in a fruit will increase with number of fruits we select. With the help of above binominal distribution probability, we can conclude that the probability of getting at most 10 fruits with black spots out of sample size of 100 is 0.035.

1.2 Confidence Interval

With confidence interval, we can determine the variations between means that is we can find the interval of true average of shelf life.

Days (Shef Life)	
Mean	6.93
Standard Error	0.28
Median	7.00
Mode	6.00
Standard Deviation	2.54
Sample Variance	6.47
Kurtosis	(0.17)
Skewness	0.25
Range	12.00
Minimum	2.00
Maximum	14.00
Sum	561.00
Count	81.00
Confidence Level(90.0%)	0.47
Lower Limit	6.46
Upper Limit	7.40

Table 2. Excel based Derivation

Parameters	Value
Mean	6.93
SD	2.54
Sample Size	81.00
Alpha	0.10
Alpha/2	0.05
T-Inv	(1.66)
Z	0.28
Z_Alpha/2	(0.47)
Confidence Level (90.0%)	0.47
Lower Limit	6.46
Upper Limit	7.40

Table 3. Manual based Derivation

Now, there are two methods for finding Confidence level,

- **CONFIDENCE NORM ()** if we know Population Standard Deviation (SD)
- **CONFIDENCE T ()** if we are estimating Population Standard Deviation from sample.

In our case we are taking **CONFIDENCE T (),** and we have entered the value in our function.

$$\textbf{Alpha} = \frac{\mathbf{100-90}\ (\boldsymbol{Confidence\ Level})}{\mathbf{100}} = 0.1$$

To find the Confidence Interval, first we want to find the Lower limit and Upper limit.

Lower Limit = Mean – Confidence Level (90%)

= 6.93 – 0.47 = 6.46.

Upper Limit = Mean + Confidence Level (90%)

= 6.93 + 0.47 = 7.40.

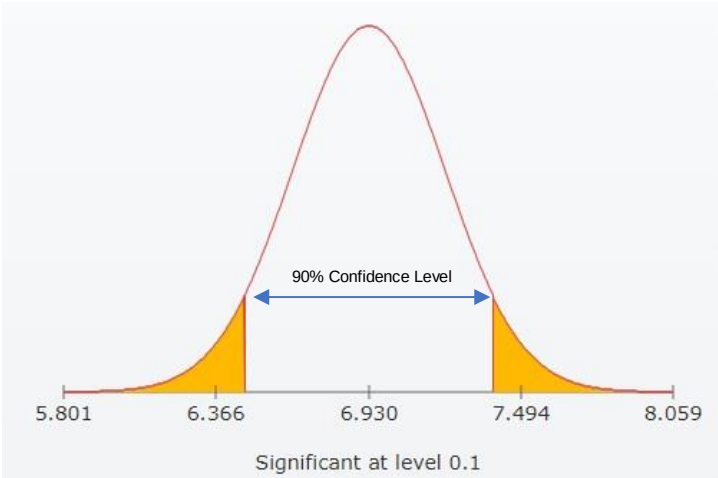

Fig 2. Confidence Level

Now, we got both Upper and Lower limit, therefore the Confidence Interval we got is 6.46 < X < 7.40, Where X represent the mean.

1.3 Hypothesis testing

1.3.1 Testing for P-values

The given Significance Level (Alpha) is 0.01. Null Hypothesis here is, the average NH4 reading of fruits with blackspot is more than that of without blackspots. So, we are going to calculate T-test assuming unequal variances because ammonium with and without blackspot is higher than 100.

t-Test: Two-Sample Assuming Unequal Variances		
	NH4 with Blackspots	*NH4 without Blackspots*
Mean	20.93	18.75
Variance	114.53	139.95
Observations	14.00	67.00
Hypothesized Mean Difference	-	
df	20.00	
t Stat	0.68	
P(T<=t) one-tail	0.25	
t Critical one-tail	2.53	
P(T<=t) two-tail	0.50	
t Critical two-tail	2.85	

H0 : μ = Average NH4 reading of fruits with blackspots is Higher
Ha : μ ≠ Average NH4 reading of fruits without blackspots is Higher

Table 4. T-test Result

As the **P(T<=t) two-tail** from the T-test two sample of unequal variance is greater than that of the sample significant value **(Alpha) 0.01**, so we are **accepting the null hypothesis** that is average NH4 of fruits with blackspots is higher than without blackspots spotting. I can conclude that I am very confident with my result.